vTrig
Indoor Evaluation Kit



User Guide

Revision 2

Vayyar Imaging Ltd.

Web: https://vayyar.com

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Chapter 1
Introduction

This guide provides information on the installation and operation of the **Vayyar Evaluation Kit.**

The kit includes HW (Vayyar three-dimensional RF-based sensor) and SW. The Vayyar sensors and operational software are used to count and track multiple people ("*targets*") in a defined space ("*arena*").

1.1 Audience

This document is targeted at the following audiences:

- Engineering and technical marketing personnel who are interested in evaluating Vayyar 3D imaging capabilities and are considering integrating Vayyar solutions into their products.

1.2 Release Information Note

Both algorithmic capabilities and features are under continuous development and improvement.

1.3 Glossary

Table 1: Glossary

Term	Definition
AC	Alternating Current
API	Application Programming Interface
DC	Direct Current
DLL	Dynamic Link Library
EVK	Evaluation Kit
GUI	Graphical User Interface
IP	Internet Protocol. An IP address is a numerical identifier assigned to a computing device or node in a TCP/IP network. The address is used to locate and identify the node in communications with other nodes on the network.
MCU	Micro-Controller Unit
PCB	Printed Circuit Board
RF	Radio Frequency
SoC	System-on-chip

Chapter 2
Installation

2.1 Required Items

Make sure you have the following items at hand when installing the system:

- Vayyar sensor
- USB cable provided with the EVK
  - Type: USB 2.0 A to micro-USB-B cable
  - One or two USB cables are provided with the kit (sensor dependent)
- PC installed with the following software:
  - Windows 10 64-bit Operating System, i7 processor is preferred
  - Chrome web browser
- EXE installer of the dedicated SW (provided by Vayyar)

2.2 Sensor Installation

The EVK sensor can be installed either on a wall or on the ceiling.

2.2.1 Wall Installation

Install the sensor on the wall and parallel to it, rotated 45° as depicted below and confirm that the USB connector is on the **right** side of the sensor (connectivity instructions to follow, see section 3.1):



Figure 1: Wall Mounting

2.2.2 Ceiling Installation

Mount the sensor on the ceiling and parallel to it so that the front of the sensor faces the floor.



Figure 2: Ceiling Mounting

2.3 Software Installation

To install the PC-based software, run the downloaded EXE file **as Administrator**.

The EXE file installer will perform the following actions:

- Installing .NET Framework and Visual C redistributable libraries
- Installing Vayyar USB drivers
- Copying the **EVK folder** to the selected path. Default path: C:\Vayyar_IndoorEVK_vX.X.X
- Creating a Desktop shortcut
- Downloading and installing Matlab runtime 9.9

NOTE

Estimated download and installation time is about 20 minutes (about 5 minutes for the EVK software, and 15 minutes for the Matlab software), depending on the computer speed and on the bandwidth of your Internet connection.

Chapter 3
System Operation

3.1 System Initialization

To initialize the system:

1. Connect the sensor to the PC as follows:
    a. Connect a USB cable to the sensor's **USB port**.
    b. Connect the other side of the USB cable to a PC-host.
2. Run the "Vayyar_IndoorEVK_vX.X.X" file – a shortcut was created on the desktop when running the installer. The "Vayyar_IndoorEVK_vX.X.X" file will:
    - Run the Vayyar EVK Engine software.
    - Run the Web GUI software using your browser (make sure it is Chrome).

➔ **The Vayyar EVK Engine**

The following command window will open:

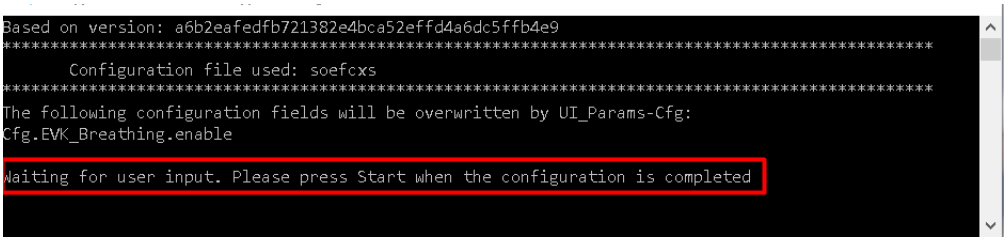


Figure 3: EVK Engine Software Running in a Command Window (CMD)

Wait for the *"Waiting for user input…"* line to appear in the command window before proceeding (as appears in the figure above). On the first operation, it takes longer, up to ~1 minute (afterwards it should take a few seconds).

➔ The Web GUI

The Web GUI welcome screen will open in an Incognito Chrome window as shown in the figure below:

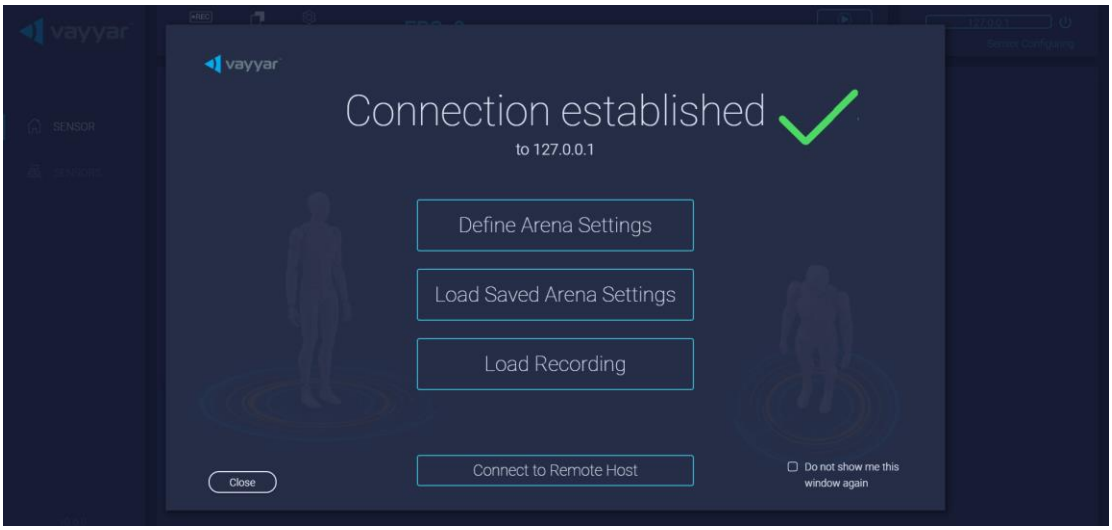


Figure 4: Web GUI Welcome Screen

3. To operate the GUI:

    a. For basic operation, click the "**Close**" button on the welcome screen; Then click the "**Run Sensor**" button on the top right corner of the screen.

    b. For more complex operation (or if you encounter any issues), follow the operating instructions in the next chapter.

Chapter 4
Operating the Graphical User Interface (GUI)

4.1 Welcome Screen

Proceed according to one of the two states of the welcome screen - “Connection established” or “Connection to Local host Failed”:

4.1.1 Connection Established

Choose one of the following options in the welcome screen:

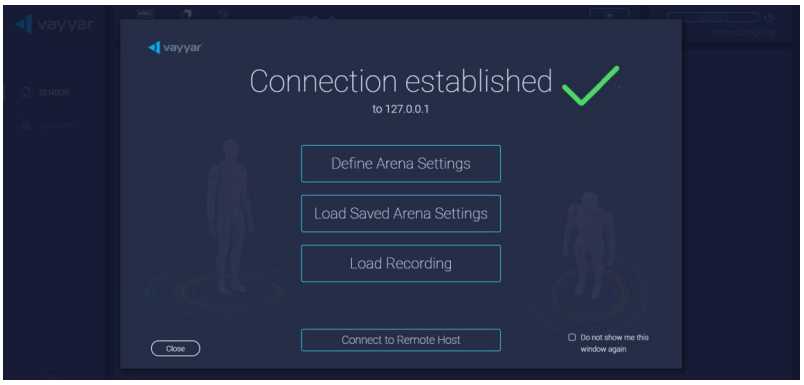


Figure 5: "Connection Established" Welcome Screen

- **Define Arena Settings** – click this button to get to the settings window, which allows the user to make a recording and configure the arena and other parameters. To learn more about the settings window, refer to section 5.1.1.
- **Load Saved Arena Settings** – click this button to load a ‘.json’ file containing previously saved arena settings.
- **Load Recording** – click this button to play and reprocess a previously saved imaging session. To learn more about loading a recording, refer to section 5.1.3.
- **Connect to Remote Host** – click this button to connect to an EVK Engine running on a different PC. To learn more about connecting to a remote host, refer to section 4.1.1.1.
- **Close** – click this button to get to the main page using the arena settings configured in the previous imaging session.
- **Do not show me this window again** – tick this checkbox to avoid the welcome screen in the next run, proceeding directly to the main page using the arena settings configured in the previous imaging session.

4.1.1.1 Connect to Remote Host

The Connect to Remote Hosts window appears as shown in the figure below.

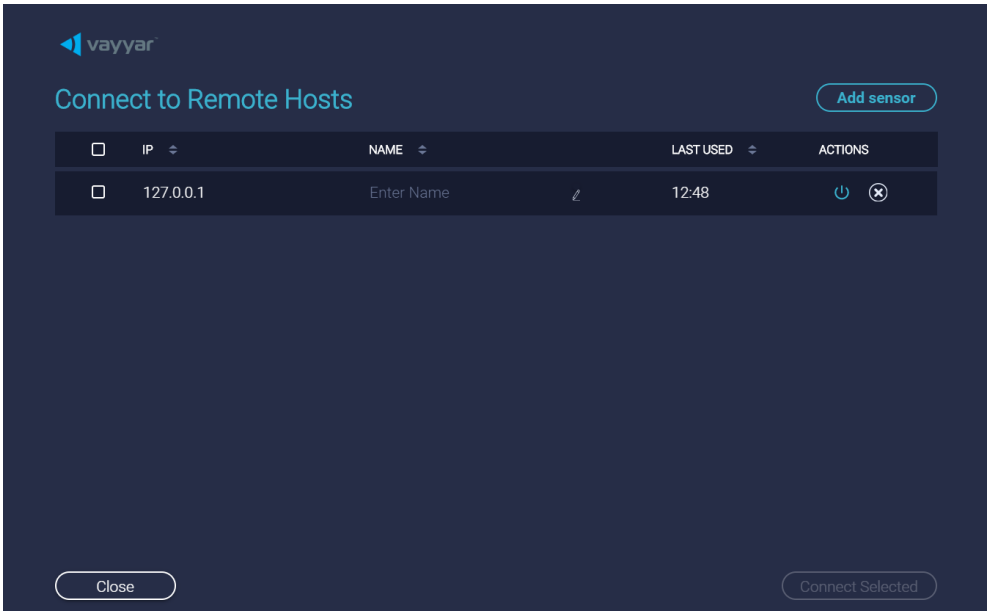


Figure 6: Connect to Remote Hosts

For connecting to an EVK Engine running on a different PC:

1. Click “Add sensor”.
2. Enter the IP address of the PC running the EVK engine.
3. Optional: enter a name for the new remote host.



Figure 7: Edit the New Remote Host

4. Click the Connect button.



Figure 8: Connect Button

4.1.2 Connection to Local Host Failed

If the Web GUI fails to connect to the EVK Engine, choose one of the following options in the welcome screen:

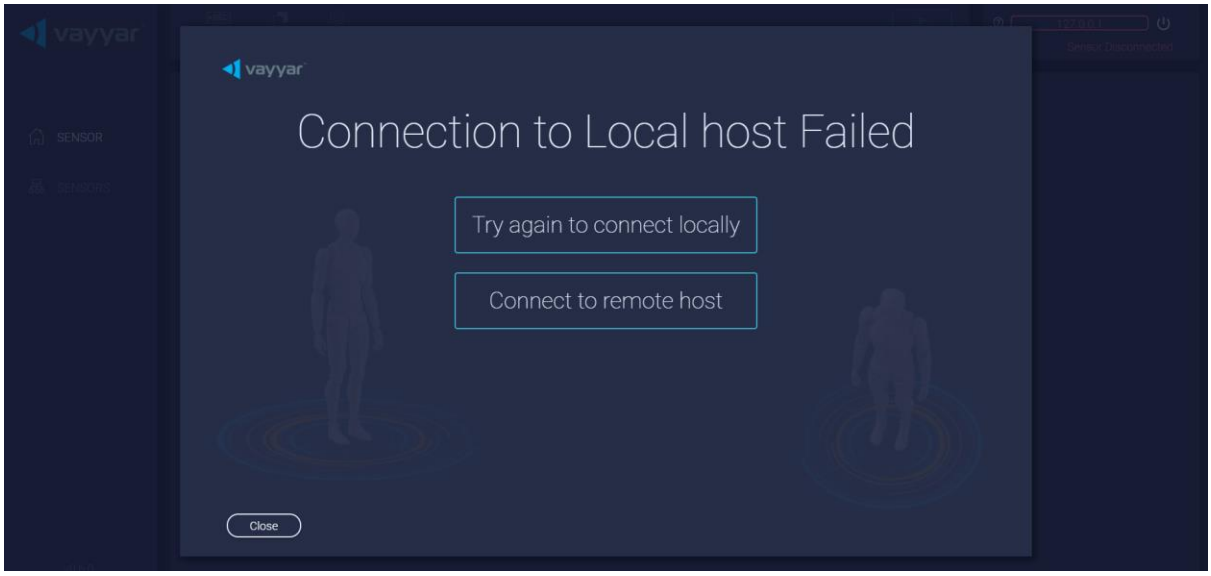


Figure 9: "Connection to Local Host Failed" Welcome Screen

➔ **Option 1 – local host**

1. Confirm that the EVK Engine (in a CMD) is still running.
2. Confirm that the EVK Engine has completed loading as shown in Figure 3 (the "Waiting for user input…" line has appeared).
3. Attempt to connect again by clicking the **Try again to connect locally** button.

➔ **Option 2 – remote host**

Connect to an EVK Engine running on a different PC by clicking the **Connect to remote host** button. To learn more about connecting to a remote host, refer to section 4.1.1.1.

Chapter 5
Main Page

The Web GUI main page is divided into the following sections:

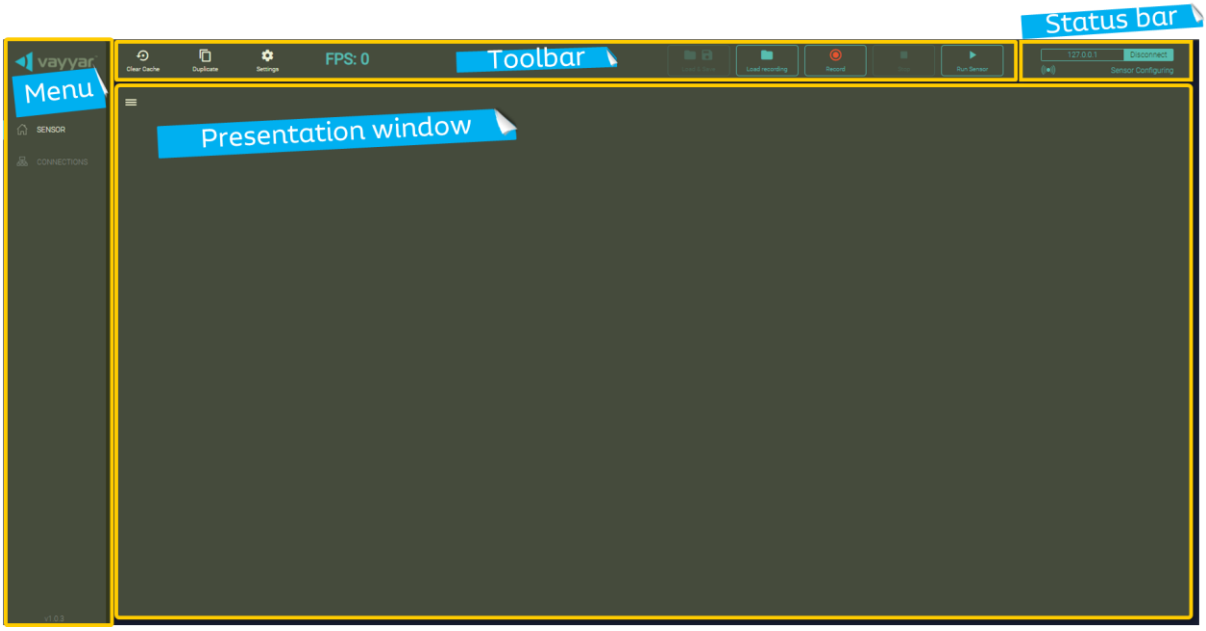


Figure 10: Web GUI Main Page Layout

- **Toolbar** - appears at the top of the page and allows the user to load a recording, control the main page windowing layout, access the settings window, and initiate the imaging session. To learn more, refer to section 5.1.
- **Status bar** - appears at the top right corner of the page and allows the user to connect and disconnect to the EVK engine (using the Power button) and specify the IP address of the PC running the EVK engine.
    - If the EVK Engine is running on the same PC as the Web GUI, leave the *localhost* address: 127.0.0.1.
- **Menu** - appears on the left side of the page. Please use the “SENSOR” tab only.
- **Presentation Window** – appears at the center of the page and presents the outputs received from the sensor. To learn more, refer to section 5.2.

5.1 Toolbar

The Toolbar appears at the top of the page as shown in the figure below.

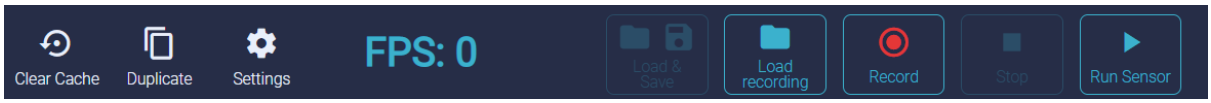


Figure 11: Toolbar

The Toolbar functions are described in the table below:

Table 2: Toolbar Functions

Function	Description
Clear Cache	Clear local storage and memory, reset settings and refresh page
Duplicate	Toggles between one presentation window and two presentation windows.
Settings	Leads to the settings window, allows the user to configure the arena and other parameters. To learn more, refer to section 5.1.1.
FPS	Specifies the frame rate of the session. To learn more, refer to section 5.1.2.
Load & Save	Enables playing a recording and save log data. To learn more about log data, refer to section 5.1.1.2.2.
Load Recording	Enables playing and reprocessing a previously saved imaging session (rawMemory data). To learn more, refer to section 5.1.3.
Record	Records an imaging session (rawMemory data). The recording can be played and reprocessed in the GUI later. To learn more about rawMemory data, refer to section 5.1.1.2.1.
Stop	Stops the imaging/recording/play recording session.
Run Sensor	Starts the imaging session.

5.1.1 Settings

The Settings window is divided into the following tabs: Arena, Save Data and Advanced – refer to the below sections.

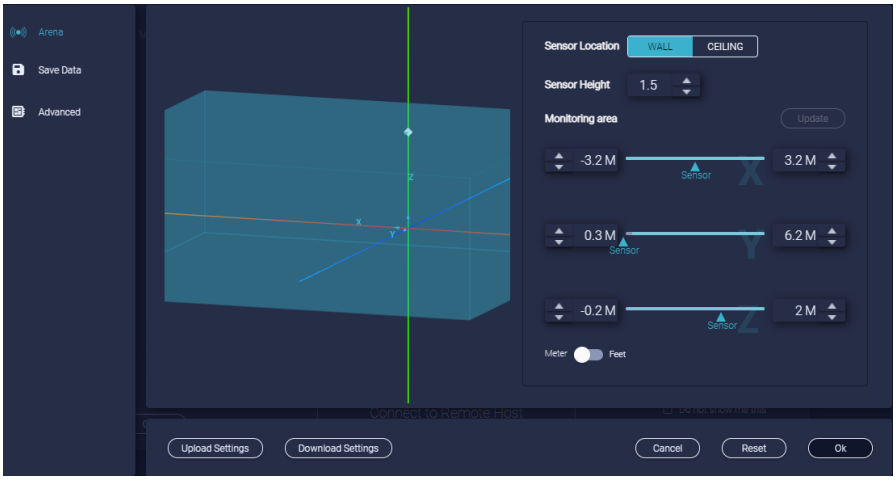


Figure 12: Settings Window

You can also perform the below actions:

- **Upload Settings** – click this button to load a '.json' file containing previously saved settings.
- **Download Settings** – click this button to save a '.json' file containing the current settings.

5.1.1.1 Arena

Arena is the area to monitor. The Arena tab appears as shown in the figure below.

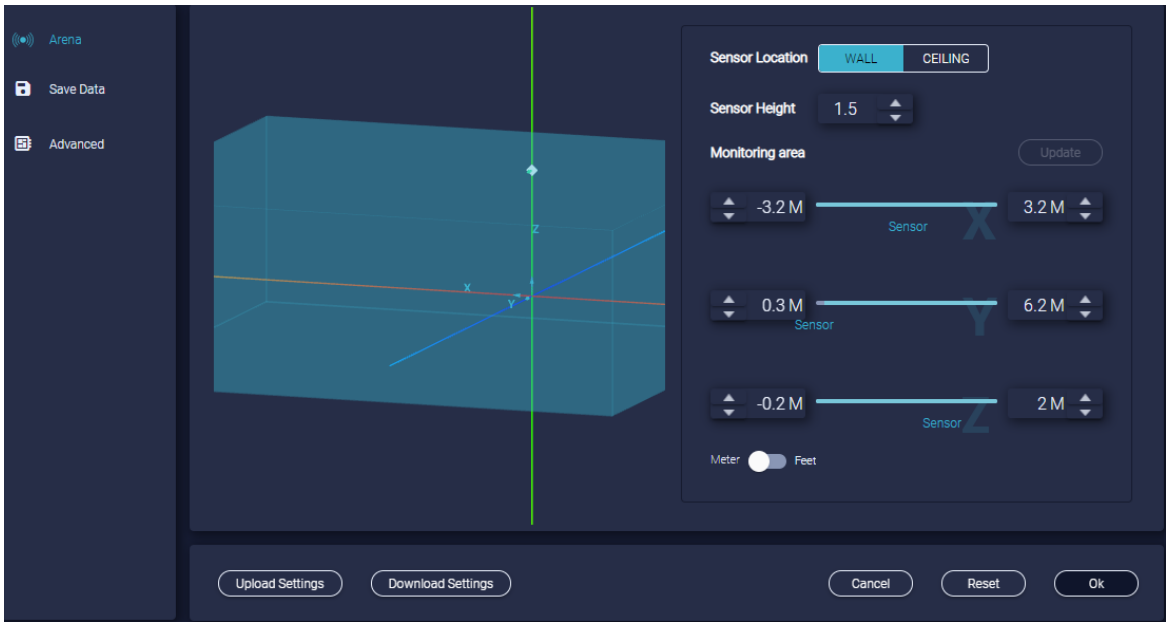


Figure 13: Arena Settings

The settings in the Arena tab are described in the table below.

NOTE

The point of origin is the point on the floor directly under the sensor. Refer to sections 5.1.1.1.1 and 5.1.1.1.2.

Table 3: Arena Settings

Setting	Description
Sensor location	The plane on which the sensor is mounted on: Wall / Ceiling.
Sensor height	The distance from the floor to the center of the sensor (meters by default).
X	The start and end points (meters by default) of the arena along the X axis, relatively to the point of origin. See orientation in sections 5.1.1.1.1 and 5.1.1.1.2.
Y	The start and end points (meters by default) of the arena along the Y axis, relatively to the point of origin. See orientation in sections 5.1.1.1.1 and 5.1.1.1.2.
Z	The start and end points (meters by default) of the arena along the Z axis, relatively to the point of origin. See orientation in sections 5.1.1.1.1 and 5.1.1.1.2.
Meter / Feet	Unit of length for the settings above: Meter or Feet.

For better performance, it is recommended to subtract the walls and ceiling from the arena by cutting 0.3m from the relevant axis.

5.1.1.1.1 Arena Axes – Wall

The arena axes for wall installation are as depicted below:

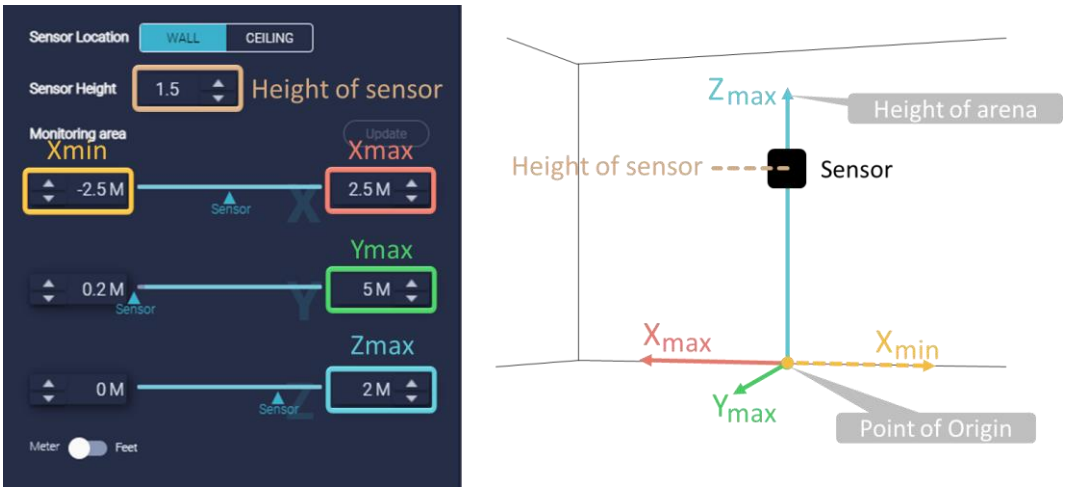


Figure 14: Arena Axes - Wall

5.1.1.1.2 Arena Axes - Ceiling

The arena axes for ceiling installation are as depicted below:

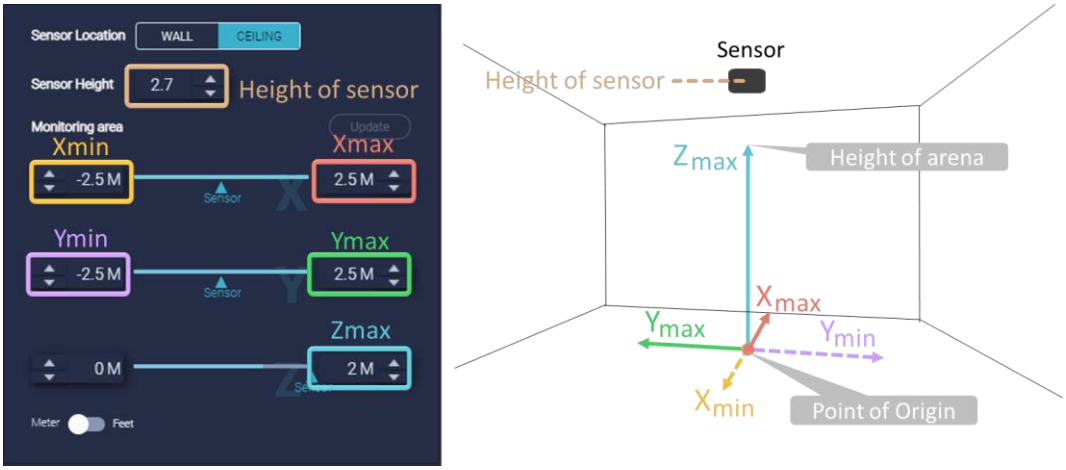


Figure 15: Arena Axes - Ceiling

The figure above relates to the following sensor orientation on the ceiling:

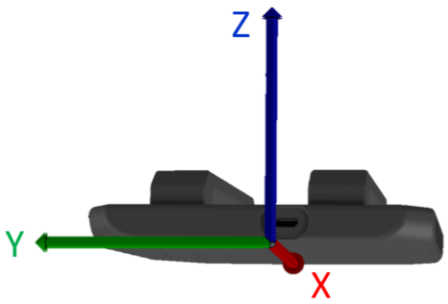


Figure 16: Ceiling Orientation

5.1.1.2 Save Data

The Save Data tab appears as shown in the figure below.

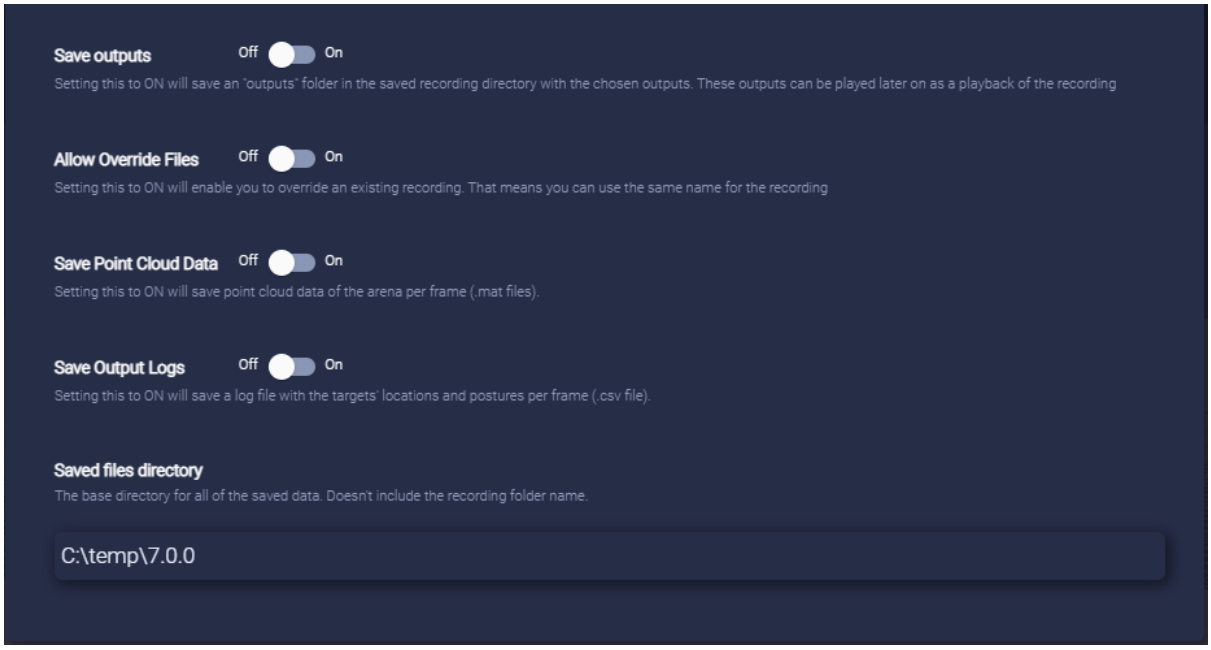


Figure 17: Save Data Settings

The settings in the Save Data tab are described in the table below.

Table 4: Save Data Settings

Setting	Description
Save outputs	**Not relevant for this EVK** – please do not use.
Allow Override Files	Turn on this switch to enable overriding the recording data in the specified path (if exists).
Save Point Cloud data	**Not relevant in this EVK** – please do not use.
Save Output Logs	Turn on this switch to save log data. To learn more, refer to section 5.1.1.2.2.
Saved files directory	The directory where the data (rawMemory and log) are saved at and played from.

5.1.1.2.1 rawMemory Data

rawMemory data is radar raw data (IQ data) in a special format. It is memory block read from the chip and depends on transmission scheme and chip architecture.

NOTE

The rawMemory data is generated for recording and playback only.

5.1.1.2.2 Log Data

.csv log file of the structure below is created per session:

Column A:	Frame Number	Number of recorded Frame
Column B:	Time Stamp	What time the frame was recorded.
From this point, every 4 columns relate to a different target.		
Column C:	Target ID	Target's ID.
Column D:	Target X	Target's location on the X axis
Column E:	Target Y	Target's location on the Y axis
Column F:	Target Z	Target's location on the Z axis

*For example, columns C – F refer to one target, columns G-J refer to the next target, Etc.

The data is saved in the path specified in the **Recordings directory** field, in a subfolder created and named according to the user input, under "ApplicationLogs".

A	B	C	D	E	F	G	H	I	J	K	L	M	N
FRAME_NUM	TIME_STAMP	TARGET_ID	TARGET_X	TARGET_Y	TARGET_Z	TARGET_ID	TARGET_X	TARGET_Y	TARGET_Z	TARGET_I	TARGET_X	TARGET_Y	TARGET_Z
18	Mar-17-2021 11:25:39.918	1	-0.13	1.28	1.83								
19	Mar-17-2021 11:25:40.140	1	-0.12	1.47	1.81								
20	Mar-17-2021 11:25:40.362	1	-0.32	1.62	1.59								
21	Mar-17-2021 11:25:40.584	1	-0.52	1.67	1.54	4	-0.23	0.51	1.32				
22	Mar-17-2021 11:25:40.807	1	-0.46	1.97	1.78	4	-0.04	0.53	1.26				
23	Mar-17-2021 11:25:41.140	1	-0.41	2.18	1.88	4	0.08	0.56	1.2				
24	Mar-17-2021 11:25:41.473	1	-0.96	2.3	1.57	4	0.12	0.5	1.15				
25	Mar-17-2021 11:25:41.695	1	-1.17	2.39	1.6	4	0.02	0.49	1.12	5	0.82	0.72	0.8
26	Mar-17-2021 11:25:41.918	1	-1.49	2.38	1.58	4	-0.02	0.49	1.2	5	0.85	0.56	0.71
27	Mar-17-2021 11:25:42.140	1	-1.66	2.41	1.56	4	0.03	0.49	1.27	5	0.67	0.49	1.04

Figure 18: Tracker Log File

5.1.1.3 Advanced

The Advanced mode appears as shown in the figure below.

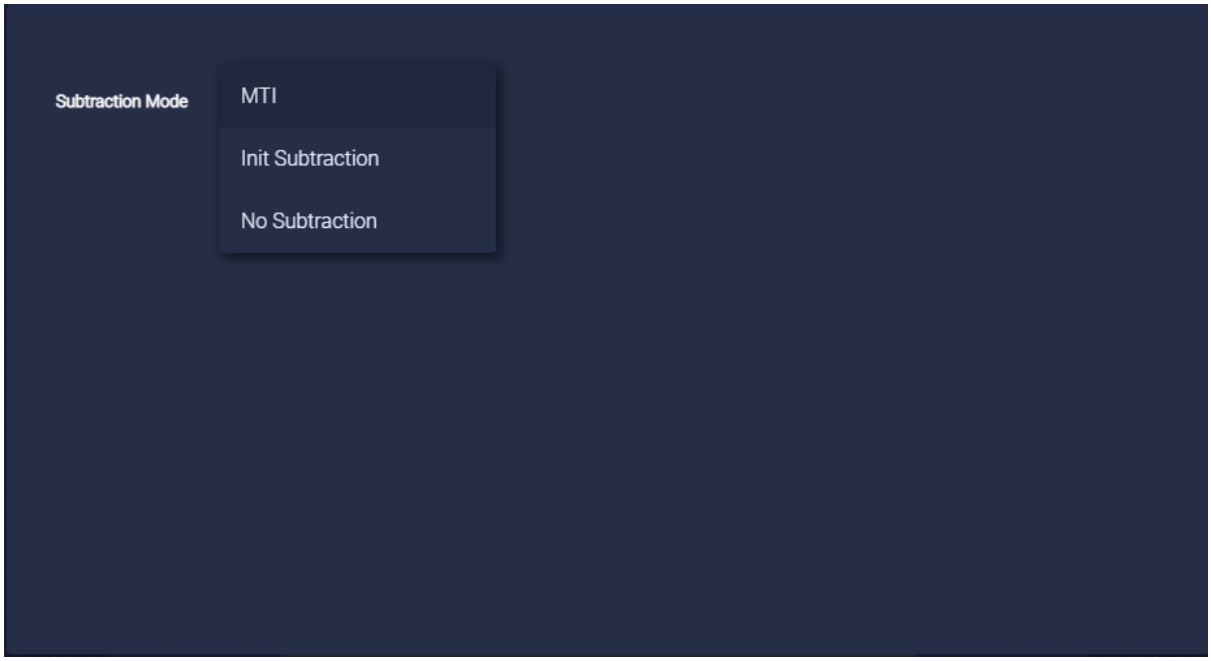


Figure 19: Advanced Mode Settings

Select the relevant work mode as described in the table below. The recommended work mode is MTI.

Table 5: Work Modes

Work Mode	Description
MTI	Shows only moving targets - displaying only the differences in the arena relative to previous few seconds of operation. This Work Mode is set as default.
Init Subtraction	Records the arena for a few seconds when sensor starts (calibration mode) and subtracts the recorded data from each future recording. It shows changes in the arena from the initial state – including moving targets (such as people) and static targets (such as chairs that moved after the calibration).
No Subtraction	Displays the arena without any subtraction.

5.1.2 FPS

The FPS (Frames per Second) label appears in the toolbar and indicates the **frame rate** of the session. The frame rate is the number of frames processed per second.

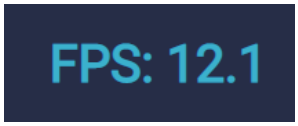


Figure 20: FPS (Frames per Second) Label

NOTE

For proper performance of the tracker and other applications, working with frame rate greater than ~8 fps is recommended. In order to increase the frame rate, you can choose stronger CPU or decrease the arena size (refer to section 5.1.1.1).

5.1.3 Load Recording

A recording is several saved *.mat* files containing raw data from the imaging session that can be reprocessed in the GUI. The recording can be reprocessed with different settings from the original ones. The result is shown in the presentation window.

To reprocess a recording:

1. Click the **Load Recording** button in the toolbar or the welcome screen.
2. Insert the recording path (including the recording name) and click Continue.

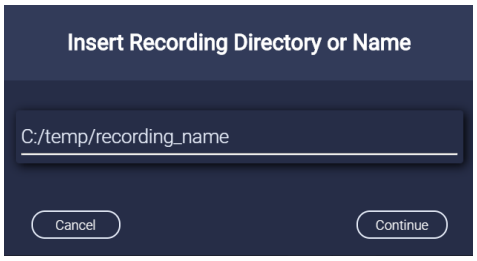


Figure 21: Insert Recording Path

3. The recording will be played in the presentation window. You can select different views to be displayed.
4. Once the recording has ended, you can click the **Load Recording** button to reprocess the same raw data again.

5.1.4 Record

To make a recording of rawMemory data:

1. Click the “Record” button on the toolbar.
2. The following menu will apepar:

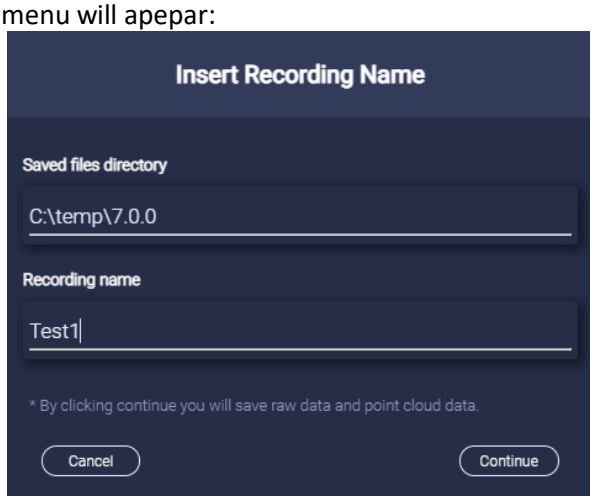


Figure 22: Record Menu

a. Insert the path to the target directory (location for saving the files) – if the "Saved files directory" is already filled in (refer to Table 4), you will not be asked to insert it again.

b. Name your recording session as you wish in the "Recording name" field.

3. Click **"Continue",** and the recording will start.

5.2 Presentation Window

The **Presentation Window** appears in the center of the page and presents the outputs received from the sensor during the imaging session.

You can choose to display either one or two presentation windows by clicking the Duplicate button in the Toolbar. Each presentation window can present a different view (to learn more about views, refer to section 5.2.1.1):

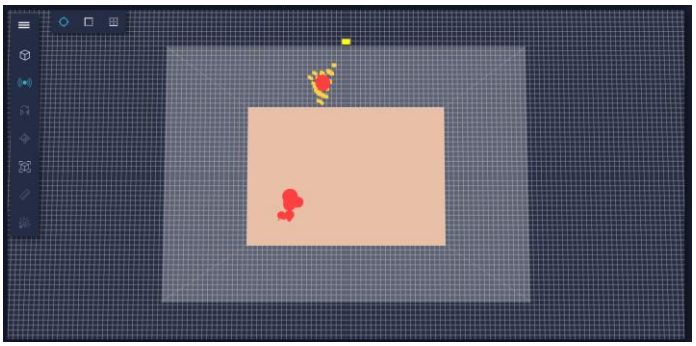


Figure 23: One Presentation Window

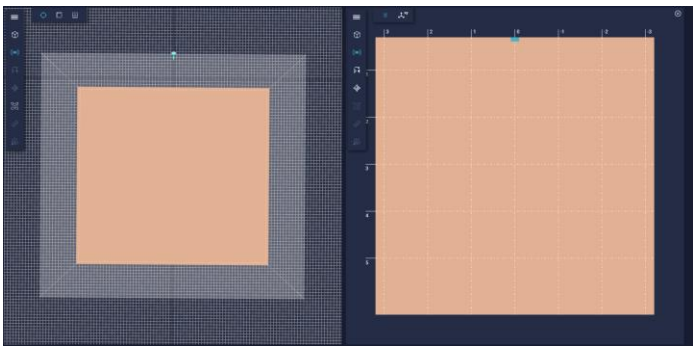


Figure 24: Two Presentation Windows

The presentation window includes two panels to control the data presentation:

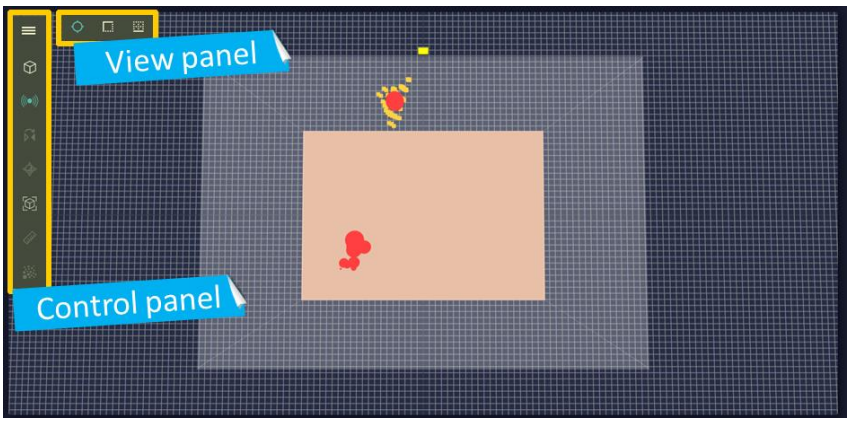


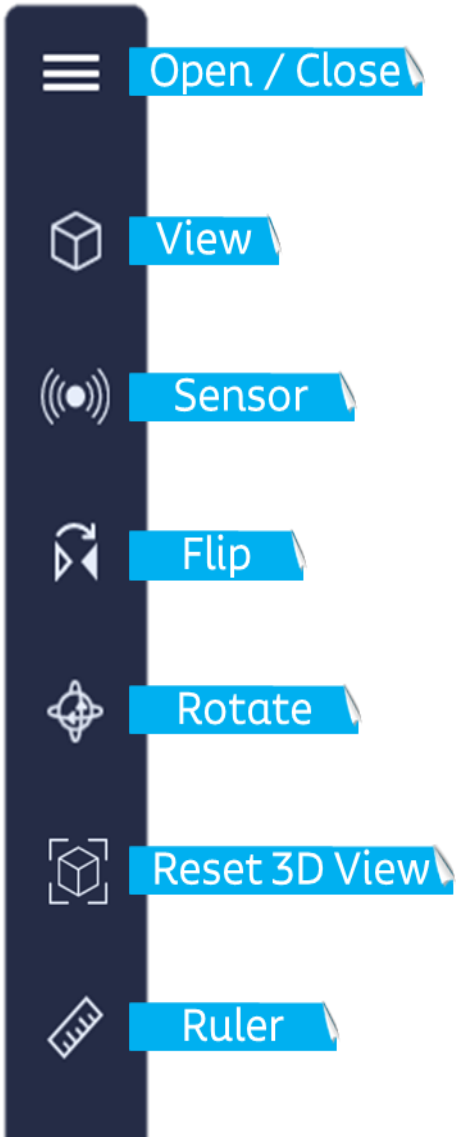
Figure 25: Panels

- **Control panel** – refer to section 5.2.1.
- **View panel** – refer to section 5.2.2.

5.2.1 Control Panel

The Control Panel's controls are described in the table below.

Table 6: Control Panel Controls

Control	Description
	Opens and closes the control panel.
	Opens a list of available views to be presented in the presentation window. Select one or more views. Refer to section 5.2.1.1.
	Toggles on and off the appearance of the sensor (visual indicator) in the arena.
	Reflects the arena horizontally. Available for 2D views only.
	Rotates the arena counterclockwise by 90° with each click. Available for 2D views only (to rotate 3D views, click and drag the arena around).
	Restores the original, default three-dimensional view. Available for 3D views only.
	Displays rulers in the arena. Available for 2D views only.

5.2.1.1 Views

The views are different presentations of the data received from the sensor.

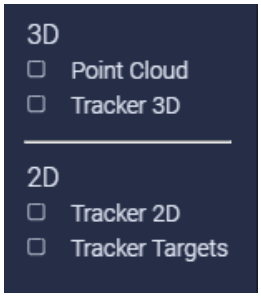


Figure 26: Views

5.2.1.1.1 3D Point Cloud

3D view of the energy received from the arena, represented as points. The colors represent intensity – center of mass will be colored red for high intensity; Edges of the body will be colored blue for low intensity (see figure below).

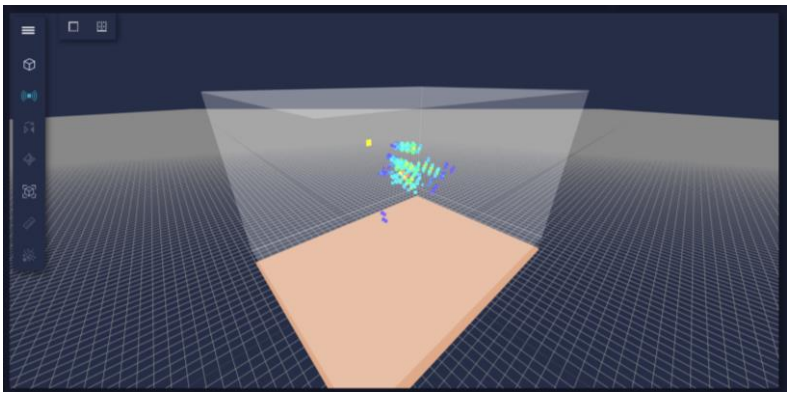


Figure 27: Point Cloud 3D Raw

5.2.1.1.2 Tracker 3D

3D view of the energy received from the arena, represented as points that are filtered and clustered by targets. Each color represents a different target. The targets' red dots represent their center. The trail of each red dot represents the target's history (locations in previous frames).

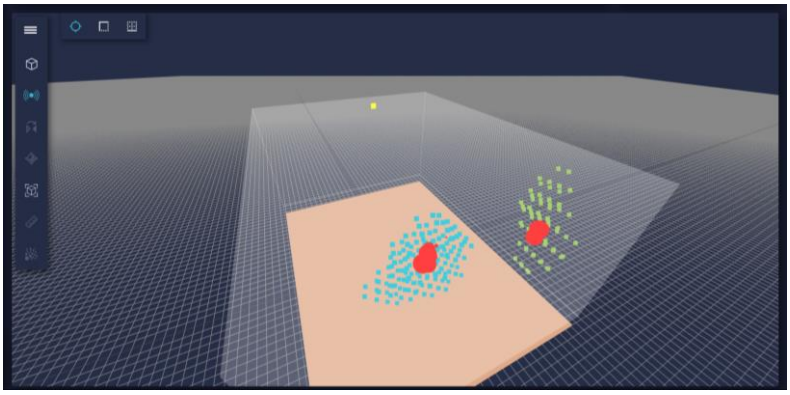


Figure 28: Point Cloud 3D Tracker

5.2.1.1.3 2D Tracker

2D view of the energy received from the arena, represented as points that are filtered and clustered by targets. Each color represents a different target. The targets' red dots represent their center. The trail of each red dot represents the target's history (locations in previous frames).

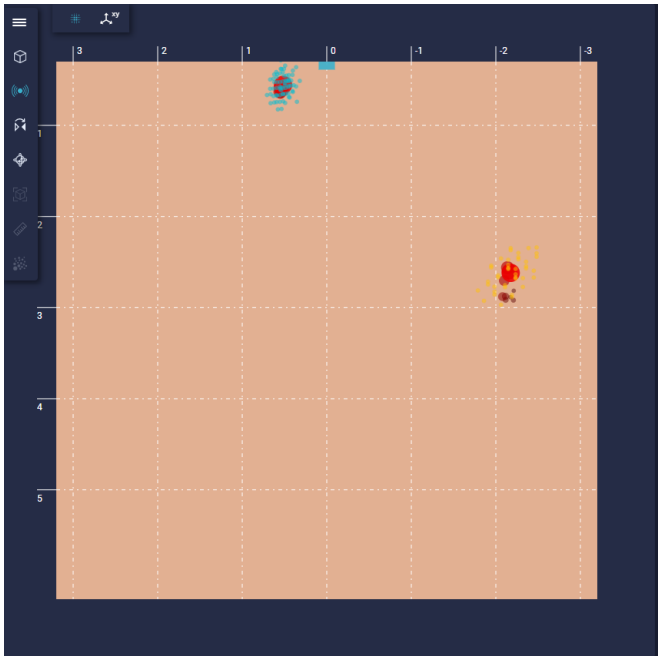


Figure 29: PointCloud 2D Tracker

5.2.1.1.4 2D Tracker Targets

Indicates the tracked targets in the arena as avatars.

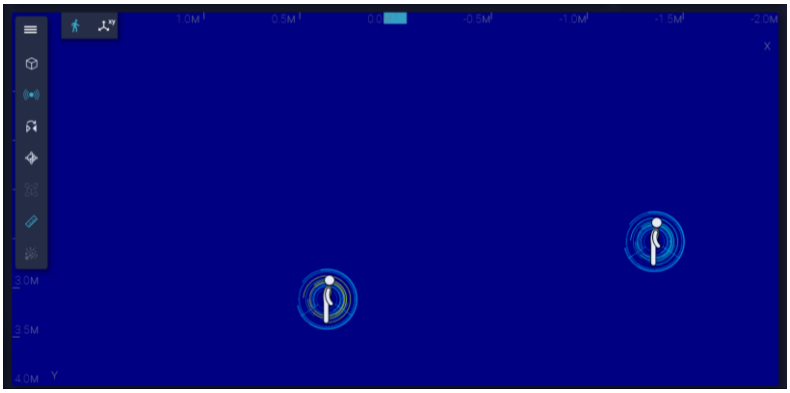


Figure 30: Tracker Targets

5.2.2 View Panel

The View Panel's controls are described in the table below. Different controls appear in the View Panel for each view.

Table 7: View Panel Controls

Control	Description
	Indicates the center of a target as a red dot.
	Presents the shadows of the targets to the "walls" of the arena.
	Presents the shadows of the targets to the "floor" of the arena.
	Presents a 2D point cloud for each target.
	Changes the 2D view axes: X-Y, X-Z or Y-Z.

END OF DOCUMENT